

Endogenous Information Acquisition in Strategic and Non-Strategic Settings

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- Tomorrow I present “Engagement Maximization” Hébert and Zhong [forthcoming]
 - We model information strategically provided by an engagement maximizing platform
- Today, I will summarize a two literatures in economics:
 - Models in which agents choose their own information to acquire
 - Often called “rational inattention”
 - Usually in a single-agent or Walrasian/large game (“non-strategic”) setting
 - Models in which a sender provides information to a receiver to persuade
 - Called “Bayesian persuasion”, inherently strategic
 - It will be useful to contrast “maximizing engagement” with “attempting to persuade”

Starting Point: States

- Let X be a finite set, so that $(X, \mathcal{F} = 2^X)$ is a measurable space.
 - Example: each $x \in X$ is a list of used cars I could buy, with varying prices and qualities
 - Using a finite X avoids some “all pain, no gain” measure theory
 - $X = \mathbb{R}^n$ with Gaussian signals only is the other popular approach
 - Some questions will be interesting when $|X| = 2$, others will need $|X| \geq 3$
- Let $\mathcal{P}(X)$ be the space of probability measures on $(X, \mathcal{F})$
 - Think of each $q \in \mathcal{P}(X)$ as an element of $\mathbb{R}_+^{|X|}$
 - with the restriction that $\iota^T \cdot q = 1$ (elements add to one)
 - Visualize: a line segment ($|X| = 2$) or an equilateral triangle (“probability simplex”, $|X| = 3$)

Starting Point: Actions and Utilities

- Let A be a finite set of actions
 - Example: buy one of the used cars on offer, or not
 - Need $|A| \geq 2$ (so there is a decision to make)
 - A few results will depend on how $|A|$ compares to $|X|$, most will not
 - The action is taken by the “agent”
- Utility function: $u : A \times X \rightarrow \mathbb{R}$
 - If agent takes action $a \in A$ in state $x \in X$, agent's payoff is $u(a, x)$
 - u encodes the agent's preference ranking over (a, x) pairs
 - If agent doesn't know X , but assigns probabilities $q \in \mathcal{P}(X)$, expected utility is

$$\sum_{x \in X} q(x) u(a, x)$$

The Indirect Utility Function

- Assume the agent chooses her action to maximize expected utility
 - Standard in economics, axioms (vN-M), but Ellsberg, Allais, etc..
- This leads to an indirect utility over probabilities $\hat{u} : \mathcal{P}(X) \rightarrow \mathbb{R}$,

$$\hat{u}(q) = \max_{a \in A} \sum_{x \in X} q(x) u(a, x).$$

- $\hat{u}$ is the utility the agent expects when choosing an action with beliefs q
- Key property: $\hat{u}$ is a convex function defined on the convex set $\mathcal{P}(X)$
 - for $q_1, q_2 \in \mathcal{P}(X)$, and $\lambda \in (0, 1)$, $q_\lambda = \lambda q_1 + (1 - \lambda) q_2 \in \mathcal{P}(X)$
 - intuition: at q_1, q_2 , can do weakly better than choosing optimal action for q_λ

- Interpretation: $\hat{u}(q)$ is “how happy am I if I act when holding beliefs q ”
- Let $\pi \in \mathcal{P}(\mathcal{P}(X))$ be a probability measure over probability measures
- Convexity gives the following property: if

$$\int \hat{u}(q') d\pi(q') = q_0,$$

then

$$\hat{u}(q_0) \leq \int \hat{u}(q') d\pi(q')$$

- this is Jensen's inequality
- Think of q_0 as “prior” (before agent learns something) and q' as “posterior” (after agent learns)

Bayes' Rule

- Now imagine the agent receives a signal $s \in S$ (another finite set)
- Suppose the agent's prior over (x, s) is $Q \in \mathcal{P}(X \times S)$
 - q_0 is the marginal distribution over x under Q
- Let $q_s \in \mathcal{P}(X)$ be the agent's posterior belief after observing $s \in S$
- Assume agent is Bayesian
 - some evidence against, e.g. Ambuehl and Li [2018]
- Bayes' rule: if $\sum_{x' \in X} Q(x', s) > 0$,

$$q_s(x) = \frac{Q(x, s)}{\sum_{x' \in X} Q(x', s)}$$

- If $\sum_{x' \in X} Q(x', s) = 0$, posterior not defined, usually arbitrary
 - In strategic settings, can matter (Ravid [2020])

Beliefs are Martingales

- The agent expects each signal with probability $\sum_{x' \in X} Q(x', s)$
- The expected value of the posterior under the prior is therefore

$$\sum_{s \in S} \sum_{x' \in X} Q(x', s) q_s(x') = q_0(x')$$

- posterior beliefs are equal in expectation to prior beliefs
 - under the agent's own expectation, not someone with different info.
 - “you cannot expect to become more optimistic”
- If we interpret the prior and posterior as occurring at different times (say, time zero and time one), then beliefs are martingales

Signals are Arbitrary

- If you give me a $\pi \in \mathcal{P}(\mathcal{P}(X))$ with finite support such that

$$\int \hat{u}(q') d\pi(q') = q_0,$$

I can make a finite set S and prior Q such that $\pi(\{q_s\}_{s \in S}) = 1$

- The signals are arbitrary: “one if by land, two if by sea” could be “two/three”, “red/green”, etc...
 - In settings where we (economists observing agents) don’t see the signals, might as well just talk about posteriors
- That is, given a prior q_0 , a “Bayes-plausible information structure” is a π satisfying “Bayes consistency” (i.e. beliefs must be martingales)

The Blackwell Ordering

- π_1 and π_2 are two information structures consistent with the same prior
- Suppose that, for each $q' \in \text{Supp}(\pi_1)$, $q' \in \text{Conv}(\text{Supp}(\pi_2))$
 - each posterior under π_1 is a convex combination of posteriors under π_2
- Then for any convex $\hat{u}$,

$$\int \hat{u}(q') d\pi_2(q') \geq \int \hat{u}(q') d\pi_1(q').$$

- i.e. π_2 is (Blackwell-) more informative than π_1
- We can reverse this logic: if for some $\hat{u}$ this holds for all such π_1, π_2 ,
 - $\hat{u}$ is convex
 - $\hat{u}$ is the indirect utility of some decision problem ($|A|$ might be infinite)

Instrumental vs. Non-Instrumental Motives

- If the agent wants information to help choose an action, her motives are “instrumental”
 - example: choose which car to buy
- If she already knows which action to take but likes information anyways, “non-instrumental”
 - example: looking at cool Ferraris you cannot buy
- As long as “more is better” (in a Blackwell-sense), $\hat{u}$ can capture both
 - knowing more about your car options is useful
 - knowing more about Ferraris is fun
- What it cannot capture: not wanting to know ex-ante
 - movie spoilers, Huntington’s disease
 - being unhappy ex-post is fine, regretting learning is not (consequence of expected utility)

Choosing an Information Structure

- Let $\Pi(q_0) \subset \mathcal{P}(\mathcal{P}(X))$ be the set of Bayes-plausible information structures
- Many models in economics with exogenous $\pi \in \Pi(q_0)$
 - when we are interested in $a(q_s)$
- Start with static endogenous info. models
 - $\pi \in \Pi(q_0)$ is chosen once, by someone
 - if agent (or someone aligned with agent), “rational inattention”
 - if not, “Bayesian persuasion”

- If agent can choose any π at no cost, w.l.o.g. will “learn x ”
 - Model collapses to full information “rational expectations”
- If agent can choose only one π , exogenous information RE
- If agent can choose from a subset $\Pi_0(q) \subseteq \Pi(q)$, more interesting
 - Agent won’t learn x , but will optimally choose π (“rational”, but “inattentive”)
- Almost equivalent: agent can choose any $\pi \in \Pi(q)$ subject to cost $C(\pi)$
 - constraints and shadow costs are equivalent via usual duality
 - but if prior/payoffs change, cost and constraint formulations differ

History of Rational Inattention

- Chris Sims started this line of work in late 90s
 - Won the Nobel prize for earlier work in empirical macroeconomics
 - RI work motivated by mismatch between macroeconomic models and data
 - Connection (1998): “Delay and inertia seems to arise even where commensurate adjustment costs are difficult to see.”
- Sims proposed $C(\pi)$ is defined by mutual information,

$$C(\pi) = \int H^S(q') d\pi(q') - H^S\left(\int q' d\pi(q')\right)$$

where H^S is the negative of Shannon's entropy (convex)

- Constraint version is $C(\pi) \leq K$, cost version subtracts $C(\pi)$ from utility

Generalized Rational Inattention

- Later work considers “posterior separable” (PS) costs,

$$C(\pi) = \int D(q', \int q' d\pi(q')) d\pi(q')$$

where $D(q_1||q_2)$ is a “divergence”

- Subset are “uniformly posterior separable” (UPS),

$$C(\pi) = \int H(q') d\pi(q') - H(\int q' d\pi(q'))$$

where H is arbitrary convex function

- Some evidence consistent with UPS, no reason to subtract $C(\pi)$ instead of $f(C(\pi))$
 - Caplin et al. [2022], Denti [2022]

Can this Model Match Data?

Caplin and Dean [2015] show:

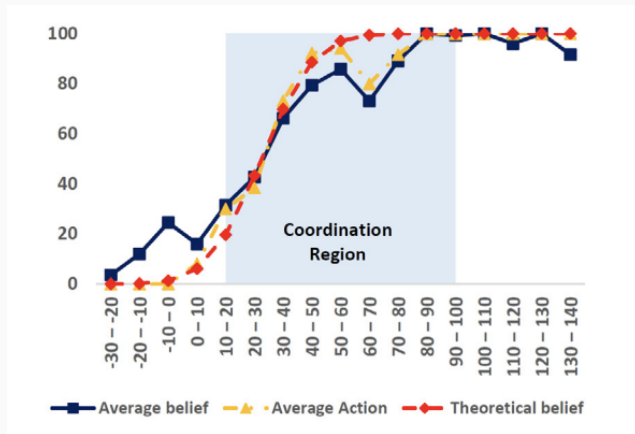
- RI models identified from “state-dependent stochastic choice” data
 - i.e. if you know true state + choice probabilities
- They do imply some restrictions
 - Loosely: “you can’t increase payoffs by swapping actions”
 - and “can’t increase payoffs by swapping signals”
 - but these are both mild and not rejected in their experimental data
- RI is a general tool to explain randomness in choice
 - They are an alternative to random utility models (Matějka et al. [2015])
- What is reject is mutual information: it doesn’t have the right prediction for how information acquisition responds to change incentives
 - maybe subtract $f(C(\pi))$ instead of $C(\pi)$?

Economics vs. Information Theory

- There is a deeper problem: Shannon's entropy is right for information, wrong for economics
 - In coding theory, the states X have no intrinsic meaning, only their frequency matters
 - In economic applications, they mean something
 - e.g. some car qualities/prices are closer to each other than others
- Ample evidence agents have a harder time discerning some states vs. others
 - Szkup and Trevino [2020], Dean and Neligh [2023]
- Hébert and Woodford [2021] develop “Neighborhood-Based Information Costs” to address this
 - closely related: “total information costs” of Bloedel and Zhong [2020]
 - both are in the “UPS” family

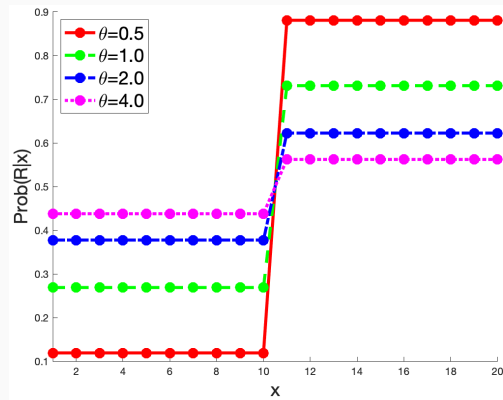
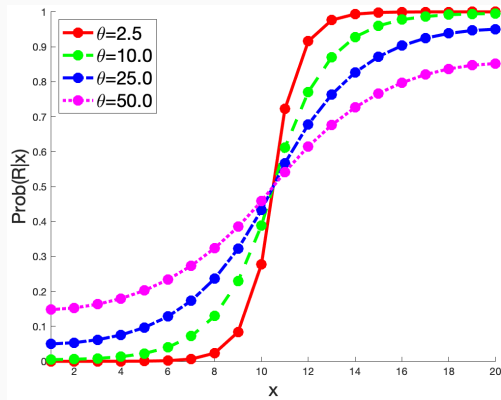
Psychometric Curves in Global Games

- ST 2020: psychometric “S” curve in laboratory play of global game



Neighborhood Cost Predicts Psychometric Curve

Neighborhood costs vs. Shannon costs in global games:



Hébert and La'O [2023] study large coordination games

- agents can learn private signals, public signals, actions of others
 - in the rational inattention way just described
- assume economy is constrained efficient under fixed info. structure
 - i.e. only possible externalities are about information choice

Results:

- Non-fundamental volatility if and only if public signals are cheaper than private research
- Inefficiency if and only watching others is cheaper than private research
- Non-fundamental volatility $\neq$ inefficient
 - monoculture does not create inefficiency; social learning does

Switching Gears: Bayesian Persuasion

- Now let's change things, and imagine the “principal” chooses $\pi \in \Pi(q_0)$
 - echoes principal/agent language; sender/receiver is standard
- Principal wants to maximize

$$\int \sum_{x \in X} q'(x) v(a^*(q'), x) d\pi(q')$$

where $a^*(q')$ is action chosen by agent

- if $v(a, x)$ is not $u(a, x)$, principal and agent may not be aligned
- usually analyzed without cost or constraint, full revelation not optimal
- Kamenica and Gentzkow [2011]

Surprising Fact: Persuasion Works

- Example: $X = \{g, b\}$ (car is good or bad), 50/50 prior
- $A = \{buy, pass\}$, $u(buy, g) = 1$, $u(buy, b) = -2$, $u(pass, \cdot) = 0$
- $v(buy, \cdot) = 1$, $v(pass, \cdot) = 0$
- No information: choose not
- But now suppose principal sends signal $s = \{rec, not\}$,

$$q_{rec}(g) = 0.75, q_{not}(g) = 0.25$$

$$q_{rec}(b) = 0.25, q_{not}(b) = 0.75$$

- Agent will now choose $a^*(q_{rec}) = buy$
 - Principal has gone from expected $v = 0$ to expected $v = \frac{1}{2}$

Tie-Breaking

- Convention: agent will do what principal wants if indifferent
 - otherwise, no optimal policies, but supremizing sequence...
- Best strategy for principal with $s = \{rec, not\}$:

$$\max_{\pi(q_{rec}), q_{rec}(g), q_{not}(g)} \pi(q_{rec})$$

subject to $q_{rec}(g) \geq \frac{2}{3}$, $\pi(q_{rec})q_{rec}(g) + (1 - \pi(q_{rec}))q_{not}(g) = \frac{1}{2}$.

- Solution: $q_{rec}(g) = \frac{2}{3}$, $q_{not}(g) = 0$,

$$\pi(q_{rec}) = \frac{3}{4}.$$

- Persuasion is surprisingly effective, but requires commitment
 - $1/4$ of the time, principal must not recommend the purchase
 - even though this will harm the principal
- Interpretations:
 - Principal hires staff/makes algorithm to make recommendations
 - Principal sets up experiment/signal, agent observes realization
 - Principal sets up experiment/signal, agent observes setup
 - Repeated game, principal reputation suffers from deviations
- Very large literature on Bayesian persuasion, including recent work relaxing commitment
 - In some settings, verifiable disclosures are sufficient

- Switch back to rational inattention, but change the model
- After choosing $\pi \in \Pi(q_0)$, receive signal, call new beliefs q_1
- Static model: choose action based on q_1
- Dynamic model: do one of two things:
 1. Act based on q_1 (same as static), or
 2. Choose another $\pi_1 \in \Pi(q_1)$, receive signal, beliefs now q_2
- Now repeat: at each time t with utility q_t , choose either:
 - stop and act, receive $\hat{u}(q_t)$
 - keep going, choose $\pi_t \in \Pi(q_t)$, beliefs update

Results without Cost of Delay

- Start with UPS cost functions:

$$C(\pi_t) = \int H(q') d\pi_t(q') - H\left(\int q' d\pi_t(q')\right)$$

- Therefore, $\mathbb{E}_0[\sum_{t=0}^{\tau} C(\pi_t)] = \mathbb{E}_0[H(q_{\tau}) - H(q_0)]$ for any stopping time τ
- The agent's problem can be written as

$$\max_{\tilde{\pi} \in \Pi(q_0)} \int (\hat{u}(q') - H(q')) d\tilde{\pi}(q') - H(q_0)$$

- With UPS cost functions and no cost of delay, the dynamics don't matter, it's identical to the static problem
 - the stopping time is also entirely indeterminate

Costs of Delay

- Let's add two costs of delay:
 - an opportunity cost of time, κ
 - an exponential discount rate ρ (assume $u(a, x) > 0$)
- If we treat $C(\cdot)$ as a cost, the optimal policy is obvious: gather all information immediately and act
 - if the time of the action is fixed at T , do nothing until T
 - “always cram right before the test”
- Instead, assume a constraint, in continuous time:

$$\lim_{h \downarrow 0} \frac{1}{h} (\mathbb{E}_{t-} [H(q_{t+h}) - H(q_{t-})]) \leq \chi.$$

Equivalent: $H(q_t) - \chi t$ is a supermartingale prior to taking an action

The Concavity of Time Preferences

- The agent's problem is now

$$\sup_{\{q_t\}, \tau} \mathbb{E}_0[e^{-\rho t} \hat{u}(q_\tau) - \int_0^\tau e^{-\rho t} \kappa dt]$$

subject to q_t is a martingale and $H(q_t) - \chi t$ is a supermartingale

- and $\{q_t\}, \tau$ in the appropriate domain (vague to limit technicality)
- Note that $e^{-\rho t}$ is a convex function of time t
 - It's better to gamble on $t = 0$ or t high vs. t medium
- Zhong [2022]: the optimal signal is one that stops immediately or conveys no information (beliefs jump, not diffuse)
 - This is not obvious from the convexity of $e^{-\rho t}$, but it is true
 - This strongest version of this result depends on UPS

- Now let's consider more general PS (not UPS) costs,

$$\lim_{h \downarrow 0} \frac{1}{h} (\mathbb{E}_{t-} [f(H(q_{t+h}) - H(q_{t-}))]) \leq \chi$$

for some strictly increasing, strictly convex f with $f(0) = 0$

- Hébert and Woodford [2023] show (with $\rho = 0, \kappa > 0$) that
 - Beliefs diffuse (follow a continuous process) in this case
 - Choices are equivalent to a static model with UPS cost based on H
 - Idea: spread out information gathering over time (“study before test”)
- with $\rho > 0, \kappa > 0$: jumps, but not necessarily immediately stopping.
 - Could be big jumps (ρ more important than $f''(\cdot)$)
 - or tiny, almost like Brownian motion ($f''(\cdot)$ more important than ρ)

Static vs. Dynamic Costs

- Bloedel and Zhong [2020] generalize this point: dynamic costs based on $D(q_1||q_2)$ can generate static costs that aren't $D(q_1||q_2)$
- In particular, if the dynamic costs are “prior-invariant”, $D(q_1||q_2) = \tilde{D}(\frac{q_1}{q_2})$, the static total cost will not be prior invariant
 - prior invariance is appealing: the cost of information shouldn't depend on one's own beliefs
 - UPS and most PS cost functions are not prior invariant
 - but this is a static intuition: different beliefs lead to different experiments, which have different costs
- Bloedel and Zhong [2020] also show that UPS costs are the unique ones that are “Sequential Learning Proof.” If you use UPS as a dynamic cost, you get UPS as the static total cost.

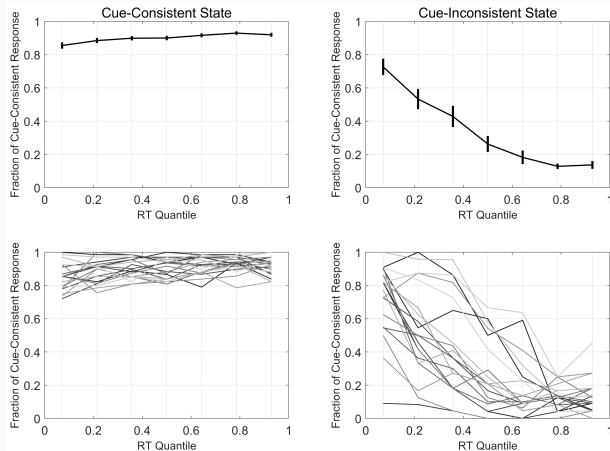
Discounting vs. Opportunity Cost

- The discount result requires strictly positive utility
 - but representation theorems don't pin down the level of u
 - resolution: there is a hidden action, "never act"
 - the relative payoffs of "act" and "never act" are changing with discounting
- Quantitatively: think about a 100,000 GBP decision, 10 hours spent thinking
 - discount at annual rate of 5% is worth ~ 6 GBP
 - opportunity cost at median hourly wage is 200 GBP
 - for lab experiments, stakes much lower, discounting matter even less
- Conclusion: in these settings, ρ does not matter much

- Hébert and Woodford [2023]: can figure out jump vs. diffusion from decision times
 - jump: step function probability of correct decision vs. time
 - diffusion: s-shaped function of correct decision vs. time
- Moreover, diffusion with total information is equivalent to DDM models of psychology
 - These were designed by psychologist to fit experimental data on response times
 - Rational inattention provides an optimizing foundation for these models
 - Why this is useful: tells you how the parameters should change in response to changing incentives/time constraints

Decisions Times by Prior

- Kelly et al 2021 data consistent with diffusion learning but not jump learning



- Rich static and dynamic literature on rational inattention
 - I haven't covered dynamic Bayesian persuasion (many papers)
- We know what happens when agents choose their own information
- We know what happens when they are being persuaded by a principal
- But what happens when they interact with a principal whose goal is engagement, not persuasion?
 - i.e. principal doesn't care what action the agent takes, as long as the agent looks at the information he provides for a long time
 - This will be the topic of my next talk

References

Sandro Ambuehl and Shengwu Li. Belief updating and the demand for information. *Games and Economic Behavior*, 109:21–39, 2018.

Alexander W Bloedel and Weijie Zhong. The cost of optimally-acquired information. *Unpublished Manuscript*, November, 2020.

Andrew Caplin and Mark Dean. Revealed preference, rational inattention, and costly information acquisition. *American Economic Review*, 105(7):2183–2203, 2015.

Andrew Caplin, Mark Dean, and John Leahy. Rationally inattentive behavior: Characterizing and generalizing shannon entropy. *Journal of Political Economy*, 130(6):1676–1715, 2022.

Mark Dean and Nathaniel Neligh. Experimental tests of rational inattention. *Journal of Political Economy*, 131(12):3415–3461, 2023.

Tommaso Denti. Posterior separable cost of information. *American Economic Review*, 112(10):3215–3259, 2022.

Benjamin Hébert and Jennifer La'O. Information acquisition, efficiency, and nonfundamental volatility. *Journal of Political Economy*, 131(10):2666–2723, 2023.

Benjamin Hébert and Michael Woodford. Neighborhood-based information costs. *American Economic Review*, 111(10):3225–55, 2021.

Benjamin Hébert and Michael Woodford. Rational inattention when decisions take time. *Journal of Economic Theory*, 208:105612, 2023.

Benjamin Hébert and Weijie Zhong. Engagement maximization. *Theoretical Economics*, forthcoming.

Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011.

Filip Matějka, Alisdair McKay, et al. Rational inattention to discrete choices: A new foundation for the multinomial logit model. *American Economic Review*, 105(1): 272–98, 2015.

Doron Ravid. Ultimatum bargaining with rational inattention. *American Economic Review*, 110(9):2948–2963, 2020.

Michal Szkup and Isabel Trevino. Sentiments, strategic uncertainty, and information

structures in coordination games. *Games and Economic Behavior*, 124:534–553, 2020.

Weijie Zhong. Optimal dynamic information acquisition. *Econometrica*, 90(4): 1537–1582, 2022.